

sol1_enhancer_grok_build

Same loop. Artifact is a **project plugin** plus three registration shims.

Grok 1.0.5 does not register skills and agents from a project plugin on its own. Symlinks are the runnable surface.

What you will build

```
.grok/plugins/ticket-enhancer/plugin.json
.grok/plugins/ticket-enhancer/agents/{enhancer-judge,enhancer-doer}.md
.grok/plugins/ticket-enhancer/skills/enhancer-loop/...

.grok/skills/enhancer-loop      -> ../plugins/ticket-enhancer/skills/enhancer-loop
.grok/agents/enhancer-judge.md  -> ../plugins/ticket-enhancer/agents/enhancer-judge.md
.grok/agents/enhancer-doer.md   -> ../plugins/ticket-enhancer/agents/enhancer-doer.md
```

Do not `grok plugin install`. That copies into `~/.grok`. Edits here then stop taking effect.

Why Grok is a different lesson

Claude discovers `.claude/` in the folder you launched from. Grok 1.0.5 discovers project plugins only after trust, and only through the shim paths.

Until trusted, `task run` finds no skill and does nothing. Headless `grok -p` never prompts. Trust is step zero.

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Learning objectives

- Build a Grok project plugin (`plugin.json` name `ticket-enhancer`)
- Register it with three `ln -sf` shims
- Trust the **git root**, not this folder
- Deny MCP writes on a read-only judge
- Keep Grok off hosted Actions

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Starting architecture

```
grok -p "/enhancer-loop --repo ..."  
└─ skill via .grok/skills/enhancer-loop (symlink)  
    spawn_subagent  
        └─ enhancer-judge allowlist + disallowedTools  
        └─ enhancer-doer  allowlist + disallowedTools
```

Subagent tool on 1.0.5 is `spawn_subagent`. A README that says `task` is wrong on this CLI.

plugin.json

```
{  
  "name": "ticket-enhancer",  
  "description": "One poll-and-act step that grooms draft tickets.",  
  "version": "0.1.0"  
}
```

`--plugin-dir` does not exist on this CLI. The shims are the install.

Trust is step zero

Trust attaches to the **git root**, not `solutions/sol1_enhancer_grok_build`.

```
task trust
grok                # accept the prompt, maybe /plugins trust
grok inspect | grep -E "enhancer-loop|enhancer-judge|enhancer-doer"
```

Check **names**, **never counts**. The Plugins line counts directories (`1 agents` even with two files, none loaded).

Allowlist plus MCP deny

```
tools:  
  - read_file  
  - grep  
  - list_dir  
disallowedTools:  
  - search_tool  
  - use_tool
```

Allowlist blocks write and spawn. Grok still injects MCP tools. Without the deny, a "read-only" judge can write an issue via GitHub MCP.

task run

```
grok --always-approve
--deny "Edit({{.LAB_ROOT}}/scripts/**)"
--deny "Edit({{.LAB_ROOT}}/work/**/tests/**)"
-p "/enhancer-loop --repo {{.TARGET}} {{.CLI_ARGS}}"
```

-p must come last.

Hooks register but never fire on 1.0.5. An empty `debug.log` does not mean the skill failed to start.

Live progress: `grok --always-approve --output-format streaming-json -p "...".`

Commands

```
cd solutions/sol1_enhancer_grok_build
cp config.json.example config.json
task clone
task trust
python3 .grok/plugins/ticket-enhancer/skills/enhancer-loop/scripts/check_fields.py --demo
task create-test-tickets
task run --
```

Expected result

Same labels and state files as Lab 1. Same marker. Same three exits.

No Grok loop skill. Polling is always external: `task poll-forever` or Actions (poor fit, see next slide).

Troubleshooting

SYMPTOM	CAUSE	FIX
Skill not found	untrusted git root	<code>task trust</code> , then inspect names
Judge wrote an issue	MCP tools injected	<code>disallowedTools: search_tool, use_tool</code>
Edits have no effect	<code>grok plugin install</code> copied away	use the shims, never install
Empty debug.log	hooks never fire on 1.0.5	stream JSON instead

GitHub Actions

Hosted Actions is a poor fit (trust prompt, local shims). Set `ENHANCER_BACKEND` to `claude`, `agent-sdk`, or `deep-agents` on the runner.

Keep Grok on a laptop or a Droplet (`labs/extra-credit/ext_5_digitalocean`).

The exits do not change if you do run it under Actions. See `labs/lab1_enhancer/GITHUB-ACTIONS.md`.

Recap

Plugin on disk. Shims to register. Trust the git root. Deny MCP writes.

Same `check_fields.py`. Same three exits. The loop is the product. The runtime is not.

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